

EDS Laboratory

Practical – 04

4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

The user should enter a list of numbers separated by space when prompted.

Output Format:

The program should display the mean of even and odd numbers separately.

Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Code:

```
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
series=pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean
```

```
grouped = series.groupby(series%2==0).mean()
# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
print("Mean of even and odd numbers:")
print(grouped)
```

Output:

✓ Test case 1 28 ms Debug	
Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd...5.0	Odd...5.0
Even...6.0	Even...6.0
dtype: float64	dtype: float64

✓ Test case 2 9 ms Debug	
Expected output	Actual output
4 4 4 6 6 2 2 2 10 12 14 16	4 4 4 6 6 2 2 2 10 12 14 16
Mean of even and odd numbers:	Mean of even and odd numbers:
Even...6.833333	Even...6.833333
dtype: float64	dtype: float64

4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

Convert the dictionary to a Pandas DataFrame.

Add a new row:

Take inputs from the user for the new row data (name, age).

Add the new row to the DataFrame.

Display the DataFrame after adding the new row.

Modify a row:

Modify a specific row by changing the age. Take the row index and new age value from the user.

Display the DataFrame after modifying the row.

Delete a row:

Take the row index to be deleted from the user.

Remove the specified row.

Display the DataFrame after deleting the row.

Add a new column:

Add a column Gender with values taken from the user.

Display the DataFrame after adding the new column.

Modify a column:

Convert names to uppercase.

Display the DataFrame after modifying the column.

Delete a column:

Remove the Age column.

Display the DataFrame after deleting the column.

Code:

```
import pandas as pd
```

```
# Provided dictionary of lists
```

```
data = {  
    'Name': ['Alice', 'Bob', 'Charlie'],  
    'Age': [25, 30, 35],  
}
```

```
# Convert the dictionary to a DataFrame
```

```
df = pd.DataFrame(data)
```

```
# Display the original DataFrame
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
name = input("New name: ")
```

```
age = int(input("New age: "))
```

```
newrow= {"Name": name,"Age":age}
```

```
df =df.append(newrow, ignore_index = True)
```

```
print("After adding a row:\n",df)
```

```
# Modifying a row
```

```
ind = int(input("Index of row to modify: "))
```

```
age = int(input("New age: "))
```

```
df.loc[ind,"Age"] = age
```

```
# Display the DataFrame after modifying a row
```

```
print("After modifying a row:")  
print(df)
```

```
# Deleting a row
```

```
ind =int(input("Index of row to delete: "))  
df = df.drop(index = ind).reset_index(drop = True)  
# Display the DataFrame after deleting a row  
print("After deleting a row:")  
print(df)
```

```
# Adding a new column
```

```
gender = input("Enter genders separated by space: ").split()  
df["Gender"]=gender  
print("After adding a new column:")  
print(df)
```

```
# Modifying a column
```

```
column = []  
for i in df["Name"]:  
    column.append(i.upper())  
df["Name"] = column  
# Display the DataFrame after modifying a column  
print("After modifying a column:")  
print(df)
```

```
# Deleting a column
```

```
df = df.drop(columns = ["Age"])
```

```
# Display the DataFrame after deleting a column
```

```
print("After deleting a column:")
```

```
print(df)
```

Output:

✓ Test case 1 94 ms   

Expected output

Original DataFrame:

.....Name..Age

0....Alice...25

1.....Bob...30

2..Charlie...35

New name: Susan

New age: 29

After adding a row:

.....Name..Age

0....Alice...25

1.....Bob...30

2..Charlie...35

3....Susan...29

Index of row to modify: 0

Actual output

Original DataFrame:

.....Name..Age

0....Alice...25

1.....Bob...30

2..Charlie...35

New name: Susan

New age: 29

After adding a row:

.....Name..Age

0....Alice...25

1.....Bob...30

2..Charlie...35

3....Susan...29

Index of row to modify: 0

New age: 27	New age: 27
After modifying a row:	After modifying a row:
.....Name..Age	Name..Age
0....Alice...27	0....Alice...27
1.....Bob...30	1.....Bob...30
2..Charlie...35	2..Charlie...35
3....Susan...29	3....Susan...29
Index of row to delete: 3	Index of row to delete: 3
After deleting a row:	After deleting a row:
.....Name..Age	Name..Age
0....Alice...27	0....Alice...27
1.....Bob...30	1.....Bob...30
2..Charlie...35	2..Charlie...35
Enter genders separated by space: F M M	Enter genders separated by space: F M M
After adding a new column:	After adding a new column:
.....Name..Age..Gender	Name..Age..Gender
0....Alice...27.....F	0....Alice...27.....F
1.....Bob...30.....M	1.....Bob...30.....M
2..Charlie...35.....M	2..Charlie...35.....M
After modifying a column:	After modifying a column:
.....Name..Age..Gender	Name..Age..Gender
0....ALICE...27.....F	0....ALICE...27.....F
1.....BOB...30.....M	1.....BOB...30.....M
2..CHARLIE...35.....M	2..CHARLIE...35.....M
After deleting a column:	After deleting a column:
.....Name..Gender	Name..Gender
0....ALICE.....F	0....ALICE.....F
1.....BOB.....M	1.....BOB.....M
2..CHARLIE.....M	2..CHARLIE.....M

4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

Display the first five rows of the data frame.

Calculate the average age of the students (limit the average age up to 2 decimal places).

Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Code:

```

import pandas as pd

# Read the text file into a DataFrame
file = input()

data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age",
"Grade"])

# write your code here..

print("First five rows:")

print(data.head())

avg_age = round(data["Age"].mean(),2)

print(f"Average age: {avg_age}")

print("Students with a grade up to B")

filtered_student = data[data["Grade"] <= "B"]

print(filtered_student)

```

Output:

✓ Test case 1 70 ms 🐞 Debug ☰ 📄	
Expected output	Actual output
studentdata.txt	studentdata.txt
First five rows:	First five rows:
<pre> ...Name...Age...Grade 0...John...25...A 1...Alin...22...B 2...Emma...24...A 3...Mike...23...C 4...Sony...21...B </pre>	<pre> ...Name...Age...Grade 0...John...25...A 1...Alin...22...B 2...Emma...24...A 3...Mike...23...C 4...Sony...21...B </pre>
Average age: 23.29	Average age: 23.29
Students with a grade up to B	Students with a grade up to B
<pre> ...Name...Age...Grade 0...John...25...A 1...Alin...22...B 2...Emma...24...A 4...Sony...21...B </pre>	<pre> ...Name...Age...Grade 0...John...25...A 1...Alin...22...B 2...Emma...24...A 4...Sony...21...B </pre>

4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by Month and calculate the total sales for each month.

Find the month with the highest total sales and display it.

Also, display the total sales for the best month.

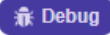
Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Code:

```
import pandas as pd
# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
df["Date"] = pd.to_datetime(df["Date"])
df["Month"] = df["Date"].dt.strftime("%Y-%m")
df["Total Sales"] = df["Quantity"] * df["Price"]
sales_by_month = df.groupby("Month")["Total Sales"].sum()
best_month = sales_by_month.idxmax()
highest_sales = sales_by_month.max()
print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```

Output:

✓ Test case 1 85 ms   	
Expected output	Actual output
sales_data.csv	sales_data.csv
Best month: 2025-01	Best month: 2025-01
Total sales: \$1210.00	Total sales: \$1210.00

4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Find the product that sold the most in terms of quantity sold.

Display the product that sold the most and the total quantity sold for that product.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Code:

```
import pandas as pd
# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
product_sales = df.groupby("Product")["Quantity"].sum()
best_product = product_sales.idxmax()
highest_quantity = product_sales.max()
# Display the result
```

```
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```

Output:

✓ Test case 1 48 ms   	
Expected output	Actual output
<code>sales_data.csv</code>	<code>sales_data.csv</code>
Best selling product: Product A	Best selling product: Product A
Total quantity sold: 23	Total quantity sold: 23

4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by City and calculate the total quantity of products sold for each city.

Find the city that sold the most products (based on the total quantity sold).

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Code:

```
import pandas as pd
# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
city_sales=df.groupby("City")["Quantity"].sum()
```

```
best_city=city_sales.idxmax()
# Display the result
print(f"City sold the most products: {best_city}")
```

Output:



4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City.

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).

Output the product pair/s that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Code:

```
import pandas as pd
from itertools import combinations
from collections import Counter
# Prompt user to input the file name
file_name = input()
# Read data from the specified CSV file
```

```

df = pd.read_csv(file_name)
group=df.groupby("Date")["Product"].apply(list)
product_pair=[]
for product_pair in group:
grouped = df.groupby("Date")["Product"].apply(list)
    product_pair = []
    for products in grouped:
if len(products)>1:
product_pair.extend(combinations(sorted(products),2))
        pair_counts =Counter(product_pair)
        if pair_counts:
max_count = max(pair_counts.values())
most_common_pairs = [pair for pair,count in pair_counts.items() if count ==
max_count]
for pair in sorted(most_common_pairs):
print(f"{pair[0]} and {pair[1]}: {max_count} times")

```

Output:

✓ Test case 1 56 ms Debug	
Expected output	Actual output
sales_data.csv	sales_data.csv
Product A and Product B: 2 times	Product A and Product B: 2 times
Product A and Product C: 2 times	Product A and Product C: 2 times

4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

Display the first 5 rows of the dataset.

Display the last 5 rows of the dataset.

Get the shape of the dataset (number of rows and columns).

Get a summary of the dataset (using .info()).

Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().

Check for missing values and display the count of missing values for each column.

Fill missing values in the 'Age' column with the median age.

Fill missing values in the 'Embarked' column with the most frequent value (mode()).

Drop the 'Cabin' column due to many missing values.

Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Code:

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())

# 2. Display the last 5 rows of the dataset
print(data.tail())

# 3. Get the shape of the dataset
print(data.shape)

# 4. Get a summary of the dataset (info)
print(data.info())

# 5. Get basic statistics of the dataset
print(data.describe())

# 6. Check for missing values
print(data.isnull().sum())

# 7. Fill missing values in the 'Age' column with the median age
data['Age'].fillna(data['Age'].median(),inplace=True)

# 8. Fill missing values in the 'Embarked' column with the mode
data['Embarked'].fillna(data['Age'].mode()[0],inplace=True)

# 9. Drop the 'Cabin' column due to many missing values
data.drop(columns=['Cabin'],inplace=True)
```

10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
data['FamilySize']=data['SibSp']+ data['Parch']

Output:

✓ Test case 1 239 ms   	
Expected output	Actual output
... PassengerId Survived Pclass ... Fare Cabin Embarked	... PassengerId Survived Pclass ... Fare Cabin Embarked
0 ... 1 ... 0 ... 3 ... 7.2500 NaN ... S	0 ... 1 ... 0 ... 3 ... 7.2500 NaN ... S
1 ... 2 ... 1 ... 1 ... 71.2833 C85 ... C	1 ... 2 ... 1 ... 1 ... 71.2833 C85 ... C
2 ... 3 ... 1 ... 3 ... 7.9250 NaN ... S	2 ... 3 ... 1 ... 3 ... 7.9250 NaN ... S
3 ... 4 ... 1 ... 1 ... 53.1000 C123 ... S	3 ... 4 ... 1 ... 1 ... 53.1000 C123 ... S
4 ... 5 ... 0 ... 3 ... 8.0500 NaN ... S	4 ... 5 ... 0 ... 3 ... 8.0500 NaN ... S
[5 rows x 12 columns]	[5 rows x 12 columns]
... PassengerId Survived Pclass ... Fare Cabin Embarked	... PassengerId Survived Pclass ... Fare Cabin Embarked
886 ... 887 ... 0 ... 2 ... 13.00 NaN ... S	886 ... 887 ... 0 ... 2 ... 13.00 NaN ... S
887 ... 888 ... 1 ... 1 ... 30.00 B42 ... S	887 ... 888 ... 1 ... 1 ... 30.00 B42 ... S
888 ... 889 ... 0 ... 3 ... 23.45 NaN ... S	888 ... 889 ... 0 ... 3 ... 23.45 NaN ... S
889 ... 890 ... 1 ... 1 ... 30.00 C148 ... C	889 ... 890 ... 1 ... 1 ... 30.00 C148 ... C
890 ... 891 ... 0 ... 3 ... 7.75 NaN ... Q	890 ... 891 ... 0 ... 3 ... 7.75 NaN ... Q

[5 rows x 12 columns]	[5 rows x 12 columns]
(891, 12)	(891, 12)
<class 'pandas.core.frame.DataFrame'>	<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890	RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):	Data columns (total 12 columns):
# Column Non-Null Count Dtype	# Column Non-Null Count Dtype
-----	-----
0 PassengerId 891 non-null int64	0 PassengerId 891 non-null int64
1 Survived 891 non-null int64	1 Survived 891 non-null int64
2 Pclass 891 non-null int64	2 Pclass 891 non-null int64
3 Name 891 non-null object	3 Name 891 non-null object
4 Sex 891 non-null object	4 Sex 891 non-null object
5 Age 714 non-null float64	5 Age 714 non-null float64
6 SibSp 891 non-null int64	6 SibSp 891 non-null int64
7 Parch 891 non-null int64	7 Parch 891 non-null int64
8 Ticket 891 non-null object	8 Ticket 891 non-null object
9 Fare 891 non-null float64	9 Fare 891 non-null float64
10 Cabin 204 non-null object	10 Cabin 204 non-null object
11 Embarked 889 non-null object	11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)	dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB	memory usage: 66.2+ KB
None	None
PassengerId Survived Pclass SibSp Parch Fare	PassengerId Survived Pclass SibSp Parch Fare
count 891.000000 891.000000 891.000000 891.000000 891.000000 891.000000	count 891.000000 891.000000 891.000000 891.000000 891.000000 891.000000
mean 446.000000 0.383838 2.308642 0.523008 0.381594 32.204208	mean 446.000000 0.383838 2.308642 0.523008 0.381594 32.204208
std 257.353842 0.486592 0.836071 1.102743 0.806057 49.693429	std 257.353842 0.486592 0.836071 1.102743 0.806057 49.693429
min 1.000000 0.000000 1.000000 0.000000 0.000000 0.000000	min 1.000000 0.000000 1.000000 0.000000 0.000000 0.000000
25% 223.500000 0.000000 2.000000 0.000000 0.000000 7.910400	25% 223.500000 0.000000 2.000000 0.000000 0.000000 7.910400
50% 446.000000 0.000000 3.000000 0.000000 0.000000 14.454200	50% 446.000000 0.000000 3.000000 0.000000 0.000000 14.454200
75% 668.500000 1.000000 3.000000 1.000000 0.000000 31.000000	75% 668.500000 1.000000 3.000000 1.000000 0.000000 31.000000
max 891.000000 1.000000 3.000000 8.000000 6.000000 512.329200	max 891.000000 1.000000 3.000000 8.000000 6.000000 512.329200
[8 rows x 7 columns]	[8 rows x 7 columns]
PassengerId 0	PassengerId 0
Survived 0	Survived 0
Pclass 0	Pclass 0
Name 0	Name 0
Sex 0	Sex 0
Age 177	Age 177
SibSp 0	SibSp 0
Parch 0	Parch 0
Ticket 0	Ticket 0
Fare 0	Fare 0

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.

Convert the 'Sex' column to numeric values (male: 0, female: 1).

One-hot encode the 'Embarked' column, dropping the first category.

Get the mean age of passengers.

Get the median fare of passengers.

Get the number of passengers by class.

Get the number of passengers by gender.

Get the number of passengers by survival status.

Calculate the survival rate of passengers.

Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below :

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
```

```
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Code:

```
import pandas as pd

import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']

# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
data['IsAlone'] = np.where(data['FamilySize']==0,1,0)

# 2. Convert 'Sex' to numeric (male: 0, female: 1)
data['Sex']=data['Sex'].map({'male':0, 'female':1})

# 3. One-hot encode the 'Embarked' column
data =pd.get_dummies(data, columns = ['Embarked'], drop_first=True)

# 4. Get the mean age of passengers
mean_age=data['Age'].mean()

print(mean_age)

# 5. Get the median fare of passengers
m =data['Fare'].median()

print(m)

# 6. Get the number of passengers by class
p =data['Pclass'].value_counts()

print(p)
```

7. Get the number of passengers by gender

```
k = data['Sex'].value_counts()
```

```
print(k)
```

8. Get the number of passengers by survival status

```
s = data['Survived'].value_counts()
```

```
print(s)
```

```
sk=data['Survived'].mean()
```

```
print(sk)
```

```
sg = data.groupby('Sex')['Survived'].mean()
```

```
print(sg)
```

Output:

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3...491	3...491
1...216	1...216
2...184	2...184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0...577	0...577
1...314	1...314
Name: Sex, dtype: int64	Name: Sex, dtype: int64
0...549	0...549
1...342	1...342
Name: Survived, dtype: int64	Name: Survived, dtype: int64
0.3838383838383838	0.3838383838383838
Sex	Sex
0...0.188908	0...0.188908
1...0.742038	1...0.742038
Name: Survived, dtype: float64	Name: Survived, dtype: float64

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

Calculate the survival rate by class.

Calculate the survival rate by embarkation location (Embarked_S).

Calculate the survival rate by family size (FamilySize).

Calculate the survival rate by being alone (IsAlone).

Get the average fare by passenger class (Pclass).

Get the average age by passenger class (Pclass).

Get the average age by survival status (Survived).

Get the average fare by survival status (Survived).

Get the number of survivors by class (Pclass).

Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
```

```
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Code:

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 1. Calculate the survival rate by class
print(data.groupby('Pclass')['Survived'].mean())

# 2. Calculate the survival rate by embarked location
print(data.groupby('Embarked_S')['Survived'].mean())

# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())

# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())

# 5. Get the average fare by class
print(data.groupby('Pclass')['Fare'].mean())

# 6. Get the average age by class
print(data.groupby('Pclass')['Age'].mean())

# 7. Get the average age by survival status
print(data.groupby("Survived")['Age'].mean())
```

8. Get the average fare by survival status

```
print(data.groupby('Survived')['Fare'].mean())
```

9. Get the number of survivors by class

```
sbc=data[data['Survived']==1]['Pclass'].value_counts()
```

```
print(sbc)
```

10. Get the number of non-survivors by class

```
nbc=data[data['Survived']==0]['Pclass'].value_counts()
```

```
print(nbc)
```

Output:

✓ Test case 1 163 ms   

Expected output	Actual output
Pclass	Pclass
1...0.629630	1...0.629630
2...0.472826	2...0.472826
3...0.242363	3...0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0...0.506073	0...0.506073
1...0.336957	1...0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0...0.303538	0...0.303538
1...0.552795	1...0.552795
2...0.578431	2...0.578431
3...0.724138	3...0.724138
4...0.200000	4...0.200000
5...0.136364	5...0.136364

6....0.333333	6....0.333333
7....0.000000	7....0.000000
10....0.000000	10....0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0....0.505650	0....0.505650
1....0.303538	1....0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1....84.154687	1....84.154687
2....20.662183	2....20.662183
3....13.675550	3....13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1....38.233441	1....38.233441
2....29.877630	2....29.877630
3....25.140620	3....25.140620
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0....30.626179	0....30.626179
1....28.343690	1....28.343690
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0....22.117887	0....22.117887
1....48.395408	1....48.395408
Name: Fare, dtype: float64	Name: Fare, dtype: float64
1....136	1....136
3....119	3....119
2....87	2....87
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
3....372	3....372
2....97	2....97
1....80	1....80

4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

Get the number of survivors by gender (Sex).

Get the number of non-survivors by gender (Sex).

Get the number of survivors by embarkation location (Embarked_S).

Get the number of non-survivors by embarkation location (Embarked_S).

Calculate the percentage of children (Age < 18) who survived.

Calculate the percentage of adults (Age >= 18) who survived.

Get the median age of survivors.

Get the median age of non-survivors.

Get the median fare of survivors.

Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below :

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Code:

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 1. Get the number of survivors by gender
sbg=data[data['Survived']==1]['Sex'].value_counts()
print(sbg)

# 2. Get the number of non-survivors by gender
nsbg=data[data['Survived']==0]['Sex'].value_counts()
print(nsbg)

# 3. Get the number of survivors by embarked location
nse=data[data['Survived']==1]['Embarked_S'].value_counts()
print(nse)

# 4. Get the number of non-survivors by embarked location
a=data[data['Survived']==0]['Embarked_S'].value_counts()
print(a)

# 5. Calculate the percentage of children (Age < 18) who survived
c=data[data['Age']<18]
print(c['Survived'].mean())

# 6. Calculate the percentage of adults (Age >= 18) who survived
a=data[data['Age']>=18]
print(a['Survived'].mean())

# 7. Get the median age of survivors
print(data[data['Survived']==1]['Age'].median())

# 8. Get the median age of non-survivors
```

```
print(data[data['Survived']==0]['Age'].median())
```

9. Get the median fare of survivors

```
print(data[data['Survived']==1]['Fare'].median())
```

10. Get the median fare of non-survivors

```
print(data[data['Survived']==0]['Fare'].median())
```

Output:

Expected output	Actual output
female...233	female...233
male....109	male....109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male....468	male....468
female....81	female....81
Name: Sex, dtype: int64	Name: Sex, dtype: int64
1...217	1...217
0...125	0...125
Name: Embarked_S, dtype: int64	Name: Embarked_S, dtype: int64
1...427	1...427
0...122	0...122
Name: Embarked_S, dtype: int64	Name: Embarked_S, dtype: int64
0.5398230088495575	0.5398230088495575
0.3810316139767055	0.3810316139767055
28.0	28.0
28.0	28.0
26.0	26.0